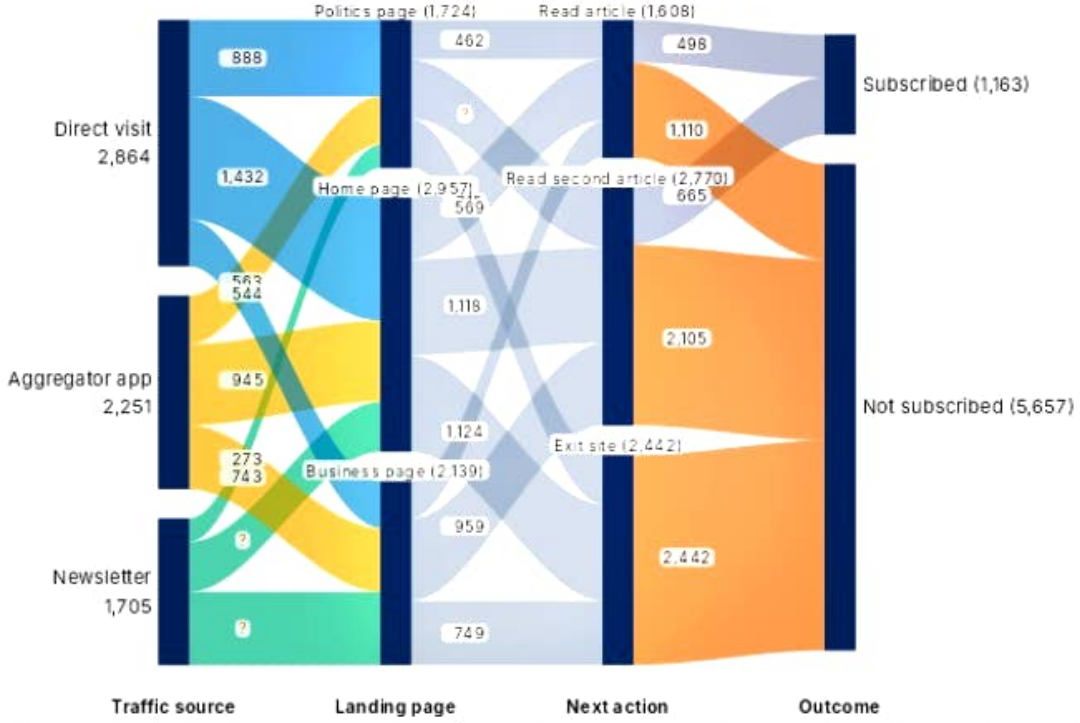
DSA06 - Data Handling and Visualization | Activity 6: Website Flow Sankey Design | Slot C

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Scenario: The analytics team of a news portal tracked 6,820 visits last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of visits on every flow. Three flows are hidden and marked "?".



(a) Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: Politics page to Read second article; Newsletter to Home page; Newsletter to Business page. State the rule you used. [4]

(b) Compute: (i) the overall conversion rate, i.e. Subscribed as a percentage of all visits; (ii) the largest single flow into "Not subscribed" or "Exit site"; (iii) the percentage of "Business page" visits that go on to "Read article". [4]

(c) Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data. [4]

(d) The Sankey is to be printed with 1 cm of node height = 300 visits. Compute the height in cm of each landing-page node (Home page, Politics page, Business page). [2]

(e) Write R code: a links data frame for the first four flows leaving "Direct visit" and "Aggregator app" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram. [4]

(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each. [2]

15/10/26

Activity -6

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Website Flow Sankey Design

a) Find the three hidden flows using conservation of flow.

Total flow entering a node is equal to the total flow leaving that node.

i) Politics page → Read Second article

Total visits to politics page = 1,724

let the missing flow be x.

$1724 = 462 + x + 569$

$x = 1724 - 462 - 569$

$\boxed{x = 693 \text{ visits}}$

ii) Newsletter → Home Page

Total visits from Newsletter = 1,705.

The outgoing flows from Newsletter are 273, 743 & the missing flow x

$1705 = 273 + 743 + x$

$x = 1705 - 273 - 743$

$\boxed{x = 689 \text{ visits}}$

iii) Newsletter → Business Page

Total visits to Business Page = 2,139

using the missing Newsletter flow calculated above:

$2139 = 959 + 431 + 749$

$\boxed{= 431 \text{ visits}}$

b) calculate the following

i) Overall Conversions rate

Total visits = 6,820

Subscribed visitors = 1,163

formula: $\text{Conversion Rate} = \frac{\text{Subscribed visitors}}{\text{Total visits}} \times 100$

$= \frac{1163}{6820} \times 100 = 17.05\%$

ii) Largest single flow into "Not Subscribed" or "Exit Site"

The flow from Exit site to Not subscribed is 2,442 visits.

iii) Percentage of Business page visitors who go to Read article

Total Business page visits = 2,139

visitors going to read article = 959

$$\text{Percentage} = \frac{959}{2139} \times 100 = 44.83\%.$$

c) Give four design decisions for the final Sankey diagram & justify each.

* Reduce Crossings: Arrange the nodes properly to reduce overlapping flows & improve readability.
* Use colours Consistently: Use different colours for traffic source or outcomes.
* Add clear labels: Display node names & visit counts so that users can understand the data.
* Highlight important flows: Highlight the 2,442 visits ending at Exit site to identify a major drop-off point.

d) Calculate the height of each landing-page node.

Given,

1 cm = 300 visits

formula: Height in cm = $\frac{\text{No. of visits}}{300}$

1) Home Page : $\frac{2957}{300} = 9.86$ cm

2) Politics Page : $\frac{1724}{300} = 5.75$ cm

3) Business page : $\frac{2139}{300} = 7.13$ cm

e) Write R code to create a Sankey diagram.

```
links <- data.frame( source = c(0,0,0,1),
                     target = c(3,4,5,3),
                     value = c(888,1432,544,543))
```

```
nodes <- data.frame(
  name = c("Direct visit",
           "Aggregator app visit",
           "Newsletter",
           "Politics Page",
           "Homepage",
           "Business page")
)
SankeyNetwork(
  Links = links,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name"
)
```

f) State two situations where a Sankey diagram becomes misleading & give one alternative chart.

1) Too many crossing flows: The diagram becomes confusing & difficult to interpret. An alternative is a bar chart.

2) Too many nodes or small flows: Labels & flows become unreadable. An alternative is a table showing the flow values.